

Running the Filter: A Verifiability × Defensibility Scorecard for Choosing Vertical Data Assets in the Post-Data-Wall Era

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Abstract

Two facts now bind frontier model development. First, the public stock of human text is finite: Epoch AI estimates an effective ~300 trillion tokens (90% CI 100T–1000T), with median full utilization near 2028 and a 2026–2032 window, and the two obvious escapes — repeating the corpus and overtraining — are closed, because repetition past ~4 epochs adds little and mechanistically damages reasoning circuitry [1,4,5]. Second, the durable replacement for scraped text is interaction with verifiable-reward environments, now the core recipe behind frontier reasoning [2,3,6]. A prior synthesis (insight i14) derived a two-axis filter for where created data compounds into a defensible asset — proprietary, hard-to-assemble data crossed with an objective, checkable outcome — but never applied it. This paper runs the filter. We first establish, with benchmark evidence (FrontierMath, SWE-bench, MedQA, MMLU, HLE), that verifiable outcomes retain measurable, ungameable headroom while static knowledge benchmarks saturate. We then score eight verticals on both axes against empirical anchors from protein folding (AlphaFold median GDT_TS 92.4 at CASP14), code, mathematics, medicine, and law, and produce a ranked shortlist: clinical claims/coding, insurance adjudication, lab/diagnostic, and contract/legal review score high on BOTH; code and competition math score high on verifiability but their public corpora thin the moat. We quantify the worked dental claims/coding example against a multi-billion-dollar, per-claim error rate. The scorecard ranks data shape only — necessary, not sufficient, for a full go/no-go.

1. Introduction

The scaling era rested on an implicit premise: that the supply of training text was effectively unbounded, so capability was a function of compute and parameters [15]. That premise is expiring on a dated schedule. Epoch AI's data-stock analysis estimates the effective stock of quality- and repetition-adjusted human public text at roughly 300 trillion tokens (90% CI ~100T–1000T), projects median full utilization around 2028 with a plausible window of 2026–2032, and notes the window pulls earlier under the overtraining that is now standard practice [1]. The two obvious escapes from this wall are closed. Repeating the finite stock does not rescue scaling: the marginal value of repeated tokens is negligible past roughly four epochs and the value of added compute decays toward zero under heavier repetition [4], and repetition does not merely fail to help — it induces a double-descent in which test loss rises mid-training and disproportionately damages the copying and induction heads that drive generalization [5].

In precisely the same window, reinforcement learning from verifiable rewards (RLVR) moved from a niche post-training trick to the core recipe behind frontier reasoning models [2]. DeepSeek-R1-Zero first demonstrated that reasoning behaviors — self-reflection, verification, strategy adaptation — can be elicited by large-scale RL on checkable signal alone, with no supervised fine-tuning, lifting AIME 2024 pass@1 from 15.6% to 71.0% [3]; process-level verifiable supervision outperforms outcome labels and majority voting, solving 78% of a representative MATH subset, with the gap widening as more solutions are sampled [6]. By late 2025 the recipe had hardened into the frontier itself: DeepSeek-V3.2 reaches parity with GPT-5 purely by scaling RL post-training

compute over a large-scale agentic-task-synthesis pipeline, and its high-compute Speciale variant surpasses GPT-5 and matches Gemini-3.0-Pro, taking gold at both the 2025 IMO and IOI [26]. Scaling the collection of verifiable environments has itself become a distinct axis: holding compute fixed, joint training across 400 procedurally generated verifiable environments lifts six-benchmark reasoning accuracy +3.37% absolute versus +0.49% for continued conventional RL at 3x the compute [27]. Together these results reframe 'environments with checkable outcomes' from a training detail into a manufacturable, fresh-signal data asset that sidesteps the human-text wall.

If verifiable environments are the next data asset class, the operational question is which verticals to build them in. Insight i14 derived a filter — proprietary or hard-to-assemble data crossed with an objective, checkable outcome — by extracting the shared template behind otherwise unrelated breakthroughs: protein folding with experimental ground truth, code with executable tests, competition math with answer keys, finance with a proprietary corpus [7,8,9,10]. But i14 stopped at the principle and never ran it. A reader is left with a two-axis rubric and no shortlist. This paper closes that gap. We (i) establish empirically that the verifiability axis has real, persistent, ungameable headroom; (ii) score eight candidate verticals on both axes against sourced evidence; (iii) produce a ranked target list and a worked high-fit example (dental claims/coding) with a quantified market signal; and (iv) bound the claim — this scorecard ranks data shape only, two of seven relevant decision factors.

2. Background and Related Work

Three literatures must be joined to motivate and run the filter. The first is the data-wall literature. Villalobos et al. (Epoch AI) forecast exhaustion of public human text between 2026 and 2032, refining an earlier estimate as data-efficiency methods advanced [1]. Muennighoff et al., training across hundreds of runs up to 900B tokens and 9B parameters, show that repetition past ~4 epochs yields negligible loss improvement and that the value of added compute decays toward zero under heavier repetition [4]; Hernandez et al. show the failure can be worse than flat — a double-descent that damages induction/copying circuitry [5]. Sorscher et al. show that high-quality pruning can beat power-law scaling, approaching exponential error decay in the pruned dataset size, making selection rather than accumulation the lever [11]. The shared implication is that the scarce good is not volume but fresh, high-coverage, learnable signal.

The second is the verifiable-reward literature. Lambert et al. (Tulu 3) name and scope RLVR to 'tasks with checkable answers,' surpassing the instruct versions of Llama 3.1, Qwen 2.5, and Mistral at comparable sizes and competing with GPT-4o-mini and Claude 3.5-Haiku [2]. DeepSeek-AI show reasoning emerging from pure RL on verifiable signal with no SFT [3]. Lightman et al. show process supervision (grading each step) solving 78% of a representative MATH subset and beating outcome supervision by a widening margin as sampling increases [6]. Benchmark work establishes that real, test-verified tasks (SWE-bench) and unpublished expert problems (FrontierMath) remained far from solved at introduction, leaving genuine headroom [8,9]; the 2025-2026 frontier has since converted much of that headroom into capability — GPT-5.2 Thinking solves 40.3% of FrontierMath (Tier 1-3, Python tools) and Claude Opus 4.5 became the first model past 80% on SWE-bench Verified at 80.9% [31,32] — precisely because the objective checker makes the gain trainable and legible. The decisive advantage in each case is supervision that can be objectively checked rather than learned from preferences, and the same property now lets the environment collection be scaled as a first-class training axis [27].

The third is the defensibility literature, which is the bridge. Model-collapse results show that recursive training on synthetic data irreversibly erases distribution tails unless real data is retained [12], making a real-data anchor a permanent dependency of the synthetic-data economy. Vertical results show domain capability tracking the domain corpus: BloombergGPT's edge from a 363-billion-token proprietary financial corpus [10], MEDITRON-70B closing

to within 5% of GPT-4 and 10% of Med-PaLM-2 via continued pretraining on a curated medical corpus [13], and Med-PaLM 2 reaching 86.5% on MedQA through medical fine-tuning [14]. Methods (LoRA, GRPO, distillation) are published and copied within months; the corpus is not. AlphaFold is the archetype that unifies all three literatures: a proprietary experimental dataset (3D-resolved structures) crossed with an objective metric, reaching a median domain GDT_TS of 92.4 at CASP14 — the first time that accuracy level was achieved — and a summed-z-score of 244.0 versus 90.8 for the next-best group [7]. Insight i14 fused these templates into a two-axis filter but never ran it; this is the gap we close.

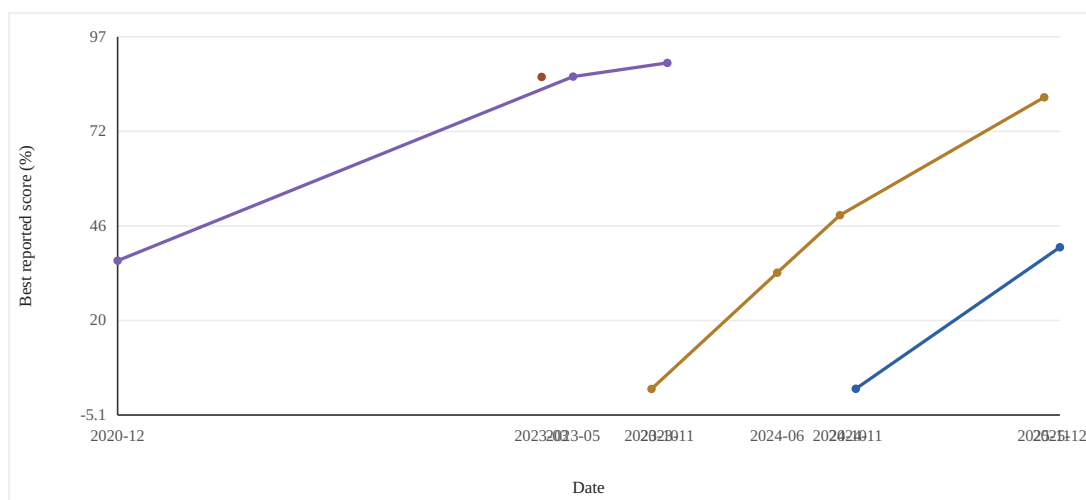
3. Methodology: Constructing and Running the Two-Axis Filter

We make the filter operational by defining each axis as an ordinal score and specifying the evidence that determines a vertical's placement. Axis one — outcome verifiability — asks whether the workflow throws off an objective, programmatically checkable pass/fail signal that can grade arbitrarily many model attempts at near-zero marginal cost. We treat verifiability as high when there exists a deterministic checker (a test suite, a numerical answer key, a payer-adjudication engine, an experimental ground truth) and low when the outcome is subjective or requires human adjudication that does not amortize. Axis two — data defensibility — asks whether the substrate that pairs with that checker is proprietary or hard to assemble (privileged clinical, contractual, or experimental records absent from pretraining) versus public (GitHub, The Stack, arXiv, competition archives). We score defensibility high when a competitor cannot reconstruct the input from the open web and low when they can.

The scoring evidence is of two kinds, and we keep them strictly separate. First, empirical anchors: published benchmark numbers and market figures, copied verbatim from primary or named sources, that establish whether each axis is satisfied for a given vertical (e.g. AlphaFold's GDT_TS for protein, SWE-bench's test-verified scores for code, MEDITRON/Med-PaLM 2 for clinical capability, LegalBench's expert-crafted task structure for legal). Second, analytical placement: the ordinal position of a vertical on the 2×2 map, which is the authors' constructed judgment from those anchors, not a measured coordinate. Figure 3 and the verifiability/defensibility ratings in Table 1 are analytical; every numeric anchor cited beside them is empirical.

To validate axis one independently of any single vertical, we assemble a benchmark time series (Section 4) contrasting verifiable, refreshable benchmarks (FrontierMath, SWE-bench, HLE) against a saturating static knowledge benchmark (MMLU). If verifiability is the renewable-signal property the filter assumes, then verifiable benchmarks should exhibit measurable, ungameable progress while non-refreshable aggregates plateau. We then run the filter across eight verticals spanning the full range of both axes — protein structure, clinical claims/coding, insurance adjudication, lab/diagnostic, contract/legal review, finance, code, and competition math — and rank them. Finally, we instantiate the top-ranked, commercially buildable vertical (clinical claims/coding, worked through its dental instance) against a quantified market signal, and we close by enumerating what the filter does NOT score so the output is not over-read.

Two data-integrity rules govern the numbers. (1) Every empirical number is reported from a named primary source — a published paper or a model developer's own reported result — rather than from unverified secondary aggregators; where a figure cannot be traced to such a source it is dropped rather than headlined. (2) Strategic-map coordinates are never presented as measurements; they are labeled analysis. These rules implement the project's standard that every empirical number trace to a real source or be dropped.



— FrontierMath (verifiable, unpublished) — SWE-bench Verified (test-verified code) — MedQA / USMLE (gradeable MCQ)
 — MMLU (saturated static knowledge benchmark)

Figure 1. Verifiable benchmarks climb while static knowledge benchmarks saturate. Benchmarks with a fresh, objectively checkable signal show steep, measurable progress (FrontierMath problems are unpublished and auto-graded; SWE-bench fixes are test-verified; MedQA is gradeable USMLE-style MCQ), while MMLU has been functionally saturated above ~86% since GPT-4's 86.4% in March 2023. FrontierMath ran from all six evaluated models under 2% at its November 2024 launch to 40.3% for GPT-5.2 Thinking (Tier 1-3, Python tools) in December 2025. SWE-bench Verified runs from Claude 2's 1.96% launch (2023) through Claude 3.5 Sonnet's 33.4% (Jun 2024) and 49.0% (Oct 2024) to Claude Opus 4.5's 80.9% (Nov 2025), the first model past 80%. The contrast is the empirical case for axis one of the filter. *FrontierMath*: Glazer et al. (arXiv:2411.04872, all evaluated models <2% at Nov-2024 launch); *GPT-5.2 Thinking* 40.3% (Tier 1-3, Python tools), OpenAI GPT-5.2, Dec 2025 (marktechpost.com/2025/12/11/openai-introduces-gpt-5-2...). *SWE-bench*: Jimenez et al. (arXiv:2310.06770, 1.96% Claude 2); Anthropic Claude 3.5 Sonnet (33.4% Jun 2024, 49.0% Oct 2024); Claude Opus 4.5 80.9% (anthropic.com/news/claude-opus-4-5, Nov 2025). *MedQA*: 36.7% (arXiv:2009.13081); *Med-PaLM 2* 86.5% (arXiv:2305.09617); *GPT-4+Medprompt* 90.2% (arXiv:2311.16452). *MMLU*: GPT-4 86.4% (GPT-4 technical report, arXiv:2303.08774); saturation per MMLU-Pro (arXiv:2406.01574) and HLE (arXiv:2501.14249). · Empirical data.

4. Evidence: Verifiability Has Real, Persistent, Ungameable Headroom

Axis one — objective, checkable outcomes — is empirically validated by the trajectory of verifiable benchmarks, which behave very differently from saturated static benchmarks. On FrontierMath, a benchmark of original, unpublished problems graded by automated answer checking, all six models evaluated scored under 2% at the November 2024 announcement [9]; a year of scaled, verifiable-reward training then moved that frontier to 40.3% (GPT-5.2 Thinking, Tier 1-3, with Python tools) [32]. This is the renewable-signal property in action: because the problems are unpublished and the answers are auto-verified, the pool resists memorization and can be refreshed and re-verified rather than leaking into training sets and decaying — and the climb is steep exactly because the checker is objective.

SWE-bench, which verifies code fixes by executing tests against real GitHub issues, tells the same story. The best system at launch, Claude 2, resolved only 1.96% of issues in 2023 [8]; Anthropic's Claude 3.5 Sonnet subsequently reported 33.4% (June 2024) and 49.0% (October 2024) on SWE-bench Verified [15], and by late 2025 Claude Opus 4.5 became the first model to clear 80% at 80.9%, ahead of Gemini 3 Pro (76.2%) [31] — a steep, sustained climb made legible and trainable by the objective pass/fail of a test suite. But the same objectivity exposes a failure mode: an audit shows state-of-the-art models locate buggy files with up to 76% accuracy from the issue text alone, versus only ~53% on repositories outside SWE-bench, evidencing that headline gains are partly memorization [28]. The field's response — contamination-resistant successors such as SWE-bench Pro, where the

best model resolves only 17.8% on a held-out proprietary commercial repository set versus ~43.6% on public repos [29] — is itself evidence both that verifiable benchmarks can be refreshed to restore discriminative power (unlike static MCQ aggregates) and that proprietary substrate is what keeps the signal hard.

Medicine confirms the pattern with a gradeable substrate: MedQA (USMLE-style) moved from a 36.7% baseline at introduction [24] to Med-PaLM 2's 86.5% via medical fine-tuning [14] and GPT-4 + Medprompt's 90.2% [25], and the static-MCQ form has since saturated — the newest frontier reasoning models (Gemini 3 Pro, GPT-5.1, GPT-5.2) now top open medical leaderboards [34]. That very saturation is why the contamination-free frontier has moved to refreshable clinical evals: on LiveMedBench, which harvests fresh real-world cases weekly to enforce temporal separation from training data, even the best of 38 LLMs reaches only 39.2% and 84% of models degrade on post-cutoff cases — direct evidence that a static medical benchmark leaks while a refreshable, rubric-graded one retains headroom [33]. Contrast all of this with a knowledge benchmark that lacks a fresh verifiable signal. MMLU has been functionally saturated above ~86% for frontier models since GPT-4's 86.4% in March 2023, with top-of-leaderboard differences statistically meaningless [16]. The field built Humanity's Last Exam precisely because LLMs now approach or exceed 90% on MMLU-style aggregates [17]; even HLE has begun to yield, with Gemini 3 Pro at 37.5% and its Deep Think variant 41.0% without tools, so the verifiable-but-refreshable benchmarks remain the only ones that keep discriminating [30]. The pattern is consistent (Figure 1, Table 2): where outcomes are objectively gradeable and the question pool can be refreshed, capability climbs measurably and the signal stays valuable; where it cannot, the benchmark saturates and stops discriminating. That is the empirical case for axis one — and the reason a vertical with a deterministic checker is a renewable data asset rather than a one-time corpus.

Table 1. Eight candidate verticals scored on data defensibility and outcome verifiability. The high-high quadrant (shortlist) pairs privileged corpora with deterministic outcomes; code and math score high on verifiability but their public corpora thin the moat (and RLVR there often elicits rather than teaches [19]). Defensibility/verifiability ratings are the authors' analytical scoring; the anchor numbers are empirical and sourced. Anchor cells for code are updated to the 2025 frontier and to the contamination-resistant SWE-bench Pro held-out commercial set (17.8%), which shows public verifiability without proprietary substrate yields a thin, replicable moat [29].

Vertical	Data defensibility	Outcome verifiability	Fit	Empirical anchor
Protein structure	High (experimental ground truth)	High (objective accuracy metric)	High (archetype)	AlphaFold median GDT_TS 92.4 at CASP14; z 244.0 vs 90.8 next-best [7]
Clinical claims/coding	High (proprietary EHR/claims)	High (CDT/ICD/X12 pass/fail)	High (shortlist)	MEDITRON-70B within 5% of GPT-4 on curated corpus [13]
Insurance adjudication	High (privileged claims data)	High (deterministic adjudication engine)	High (shortlist)	Initial denial rate 11.8% in 2024 [22]
Lab/diagnostic workflows	High (proprietary test records)	High (confirmed test outcome)	High (shortlist)	Med-PaLM 2 86.5% MedQA via clinical fine-tuning [14]
Contract/legal review	High (privileged corpora)	High (checkable clause outcomes)	High (shortlist)	LegalBench: 162 expert-crafted tasks, 6 reasoning types [18]
Finance	High (proprietary corpus)	Medium (less cleanly pass/fail)	Medium-high	BloombergGPT 363B-token proprietary corpus [10]
Code	Low (public: GitHub / The Stack)	High (tests pass)	Medium (thin moat)	SWE-bench public 1.96% (Claude 2, 2023) -> 80.9% (Claude Opus 4.5, 2025) [8,31]; held-out commercial set only 17.8% [29]
Competition math	Low (public archives / arXiv)	High (answers check)	Medium (thin moat)	FrontierMath: all evaluated models <2% at Nov-2024 launch [9]

Per-row citations as listed; see References. AlphaFold GDT_TS and z-scores from Jumper et al. (Nature 2021) / CASP14 assessment.
Verifiability/defensibility ratings are analytical scoring of the cited evidence.

Table 2. Benchmarks with a fresh, objectively checkable signal (FrontierMath, SWE-bench, MedQA, HLE) show measurable progress and retained discriminative power; the static, non-refreshed aggregate (MMLU) saturates. This validates axis one of the filter: objectively checkable, refreshable outcomes keep producing usable gradient signal, while static datasets decay. Figures are reported from primary sources; later secondary-aggregator frontier reads are not headlined. Latest cells are now populated with primary-source-traceable 2025-2026 frontier reads (GPT-5.2, Claude Opus 4.5, Gemini 3); contamination-resistant variants (SWE-bench Pro) and refreshable evals (LiveMedBench) confirm that the discriminative power survives only where the signal is verifiable and refreshable.

Benchmark	Verifiable / refreshable?	Earliest reported	Latest reported	Status
FrontierMath	Yes (unpublished, auto-graded)	<2% (all six evaluated models, Nov 2024) [9]	40.3% GPT-5.2 Thinking (Tier 1-3, Python tools), Dec 2025 [32]	Open headroom
SWE-bench Verified	Yes (test-verified execution)	1.96% Claude 2 (2023) [8]	80.9% Claude Opus 4.5 (Nov 2025), first model past 80% [31]	Climbing; contamination-resistant Pro variant restores headroom (commercial set 17.8%) [29]
MedQA (USMLE)	Yes (gradeable MCQ)	36.7% baseline (2020) [24]	90.2% GPT-4+Medprompt [25]	Saturated; frontier reasoning models now top medical leaderboards [34]
MMLU	Partially (static MCQ, not refreshed)	86.4% GPT-4 (Mar 2023) [16]	Frontier models near or above ~90% [17]	Saturated
Humanity's Last Exam	Yes (expert, refreshable)	2.7-21.6% at launch (early 2025) [17]	37.5% Gemini 3 Pro / 41.0% Gemini 3 Deep Think (no tools), Nov 2025 [30]	Open headroom; still discriminating at the frontier [30]

FrontierMath: arXiv:2411.04872 (all evaluated models <2% at Nov-2024 launch). SWE-bench: arXiv:2310.06770 (Claude 2 1.96%); Anthropic Claude 3.5 Sonnet (33.4% Jun 2024, 49.0% Oct 2024). MedQA: arXiv:2009.13081 (36.7%); arXiv:2305.09617 (Med-PaLM 2 86.5%); arXiv:2311.16452 (GPT-4+Medprompt 90.2%). MMLU: GPT-4 technical report arXiv:2303.08774 (86.4%); MMLU-Pro (arXiv:2406.01574). HLE: arXiv:2501.14249 (2.7-21.6% at launch). 2025-2026 frontier reads: GPT-5.2 FrontierMath 40.3% (marktechpost, Dec 2025); Claude Opus 4.5 SWE-bench Verified 80.9% (anthropic.com/news/claude-opus-4-5, Nov 2025); Gemini 3 Pro/Deep Think HLE 37.5%/41.0% (blog.google/products/gemini/gemini-3, Nov 2025); medical leaderboard leaders per Medmarks (arXiv:2605.01417); contamination-free clinical eval per LiveMedBench (arXiv:2602.10367); SWE-bench Pro commercial-set 17.8% (arXiv:2509.16941).

5. Analysis: Running the Filter Across Eight Verticals

AlphaFold anchors the high-high corner and calibrates both axes: a proprietary experimental dataset (3D-resolved protein structures) crossed with an objective accuracy metric, reaching median GDT_TS 92.4 at CASP14 and outperforming all other groups by a summed-z margin of 244.0 to 90.8 [7]. It is the existence proof that proprietary substrate × checkable outcome produces a capability competitors cannot copy without the same private data.

Code and competition mathematics score maximally on verifiability — tests that pass, answers that check [8,9] — but their corpora are largely public: GitHub, The Stack, arXiv, competition archives. Public substrate thins the moat, because any competitor can assemble the same input and the only durable differentiator becomes method, which commoditizes within months. A further, sharper caveat applies: RLVR on these substrates frequently elicits capability the base model already latently holds — gains appear on Qwen math even from spurious or random rewards, yet fail to transfer to Llama or OLMo [19] — so created data in code/math often reveals rather than teaches. These verticals are therefore medium-fit: excellent for demonstrating verifiable-reward technique, weak as a defensible data asset.

The high-high quadrant for buildable commercial verticals is occupied by workflows that pair a private clinical or contractual record with a deterministic outcome. Clinical claims/coding adjudication pairs proprietary EHR/claims data (axis two) with CDT/ICD/X12 codes that deterministically pass or fail a payer's adjudication engine (axis one). Insurance adjudication, lab/diagnostic test workflows, and contract/legal review share the shape: privileged corpora crossed with checkable clause, coding, or test outcomes [13,14,18]. LegalBench — 162 tasks across six types of legal reasoning, hand-crafted by legal professionals [18] — evidences that legal outcomes can be expert-defined and checkable (axis two) over corpora that are typically privileged (axis one). Medicine evidences that the proprietary clinical corpus is what drives expert-level capability and closes the gap to frontier models [13,14], confirming both axes simultaneously. Finance sits high on defensibility via BloombergGPT's proprietary 363B-token corpus [10] but lower on verifiability, since its outcomes are less cleanly pass/fail than code, math, or coding adjudication — placing it just outside the buildable high-high shortlist.

The worked high-fit example is dental claims/coding: a private clinical record paired with a deterministic coding/claims outcome. It scores high on both axes by construction — the record is proprietary to the clinic, and the CDT code either adjudicates or denies against the payer's engine. Critically, it also satisfies the 'teaches rather than reveals' criterion: clinical coding workflows are under-documented in pretraining, so created verifiable-reward data adds genuine capability rather than surfacing latent skill [19]. This is the canonical instance the filter nominates, and it is the worked example we quantify in Section 7.

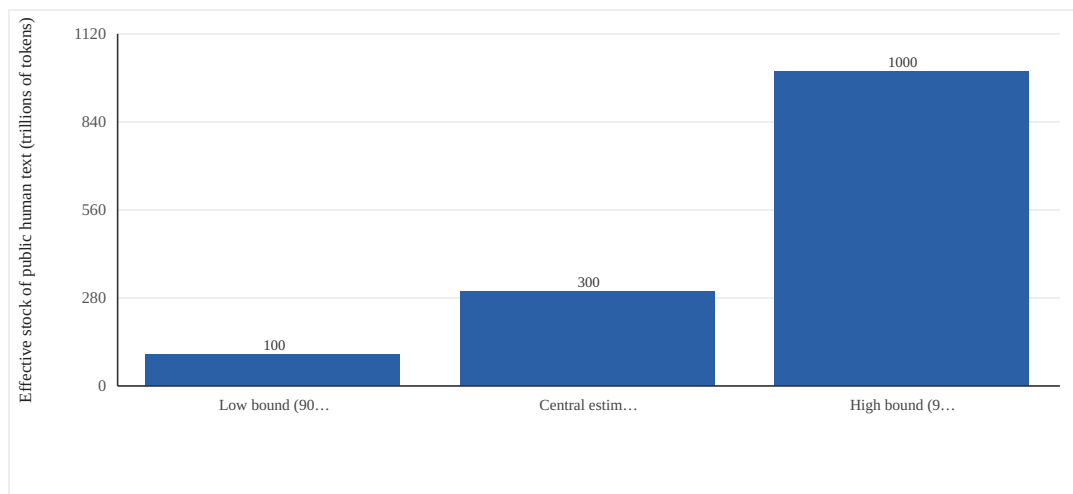


Figure 2. The human-text data wall: a fixed ~300T stock against a 2026-2032 exhaustion window. Epoch AI estimates the effective stock of public human-generated text at ~300 trillion tokens (90% CI 100T-1000T), projected to be fully utilized between 2026 and 2032 under compute-optimal training (median ~2028; sooner under aggressive overtraining). The stock is fixed and its CI bounds are exact-sourced; we plot only those three sourced values and do not fabricate a per-year demand curve, since Epoch does not publish clean per-year token-demand points. Supply flattening coincides with the marginal token's value bending negative past ~4 epochs of repetition, which is why fresh verifiable signal (the renewable alternative) becomes the scarce good. Villalobos et al., 'Will we run out of data? Limits of LLM scaling based on human-generated data,' Epoch AI / ICML 2024 (arXiv:2211.04325). Stock ~300T tokens (90% CI 100T-1000T), median ~2028, window 2026-2032. · Empirical data.

6. Implications for Data Strategy

First, prioritize the high-high quadrant: clinical claims/coding, insurance adjudication, lab/diagnostic, and contract/legal review. These are the verticals where created verifiable-reward data compounds into an asset

competitors structurally cannot assemble, because the input is privileged and the outcome is deterministic. The filter converts a principle into a concrete front-of-funnel.

Second, treat code and math as capability demonstrations, not as the asset. Their public corpora mean the data moat is thin even though the verifiable signal is excellent, and RLVR there often elicits rather than teaches [19]. Building RLVR competence on code/math is sound R&D; mistaking it for a defensible data business is a category error the filter exposes.

Third, recognize that the highest-value verifiable data lies where base models lack latent skill — specialized, under-documented verticals absent from pretraining, precisely the proprietary clinical and contractual workflows the scorecard ranks highest [19]. There the created data teaches; the marginal token is worth most exactly where the open web is thinnest.

Fourth, pair every training-data offering with a continuously refreshed, contamination-free evaluation in the same vertical. Benchmarks decay: MMLU saturated, named datasets leak, and the contamination-free response is a rolling, held-out question stream [16,17]. The evidence that this is mandatory rather than optional is now quantified — SWE-bench gains are partly memorization (76% in-benchmark vs ~53% out-of-benchmark file-path identification) [28], and 84% of LLMs degrade on the post-cutoff cases of a contamination-free clinical benchmark [33]. A verifiable environment supports both an unbounded training-rollout stream and a regenerable eval stream from the same substrate — and the eval side carries a built-in renewal motive, since any static eval is contaminated once it leaks into a training set. The same asset thus serves two markets and resists the commoditization that hits static datasets.

Fifth, price curation as compute-relative and recurring, not as a frozen dataset. The optimal corpus is a moving target across compute budget, inference demand, and deployment precision; a verifiable environment re-derives fresh training-and-eval signal on demand, which is a subscription surface a one-time data dump cannot capture.

7. Investment Thesis

The investable claim is timing plus defensibility. Timing: the public-text data wall is not a distant abstraction but a dated forecast — ~300T effective tokens, median exhaustion ~2028, window 2026–2032, pulled earlier under standard overtraining [1] — and the replacement signal (verifiable environments) is no longer emerging but the frontier recipe itself: scaling RL over agentic-task-synthesis pipelines now reaches GPT-5/Gemini-3-Pro parity [2,3,26], and scaling the environment collection is its own capability axis [27]. Capital deployed now to manufacture proprietary verifiable data buys into the asset class precisely as the substitute (scraped text) runs out and demand for fresh gradient signal inflects upward. This is a market in formation, not a mature one.

Defensibility: the scorecard isolates where that data becomes a moat rather than a commodity. The lesson of code and math is the cautionary one — high verifiability over public data is not defensible, because the input is replicable and method commoditizes within months [8,9]. The lesson of AlphaFold, BloombergGPT, and the clinical corpora is the bankable one — proprietary input crossed with a checkable outcome produces capability competitors cannot copy without the same private data [7,10,13]. An investor should fund the quadrant, not the technique. And the economics are software-shaped: the marginal unit of training signal from a built verifier is a program execution, not a human label, so gross margin scales with usage rather than headcount.

The market signal under the top-ranked vertical is concrete and large. Medical coding errors cost the U.S. healthcare system on the order of \$36 billion annually in lost revenue, denied claims, and penalties [20,21]; the American Medical Association estimates roughly 12% of medical claims are submitted with inaccurate codes [21];

initial denial rates rose to 11.8% in 2024 (up from 10.2%) [22]; the cost to rework a single denied claim runs \$40–\$118 [20]; and Premier estimates providers spent on the order of \$19.7 billion in 2022 fighting denials, with later survey reads putting claims-adjudication costs at ~\$25.7 billion [20,22]. Each of these is a per-claim, objectively adjudicated outcome — a verifiable reward signal attached to a multi-billion-dollar error rate, sitting on top of proprietary clinical records. That is the rare combination the filter is designed to find: a deterministic outcome, a privileged corpus, and a quantified willingness-to-pay anchored in real, recurring losses.

Finally, deployment economics favor the thesis. Inference-aware and precision scaling are fragmenting models into many small specialists, and each specialist is a distinct customer for a distinct curated vertical dataset and eval [23]. The buyer base is not a handful of frontier labs but a proliferating field of vertical model-builders — a structural demand multiplier that rewards a productized, repeatable data-and-eval offering over bespoke one-offs. The bet is to own the real-data anchor in a vertical the open web underrepresents but whose outcomes are deterministically checkable; clinical claims/coding, worked through the dental instance, is the archetypal high-fit target.

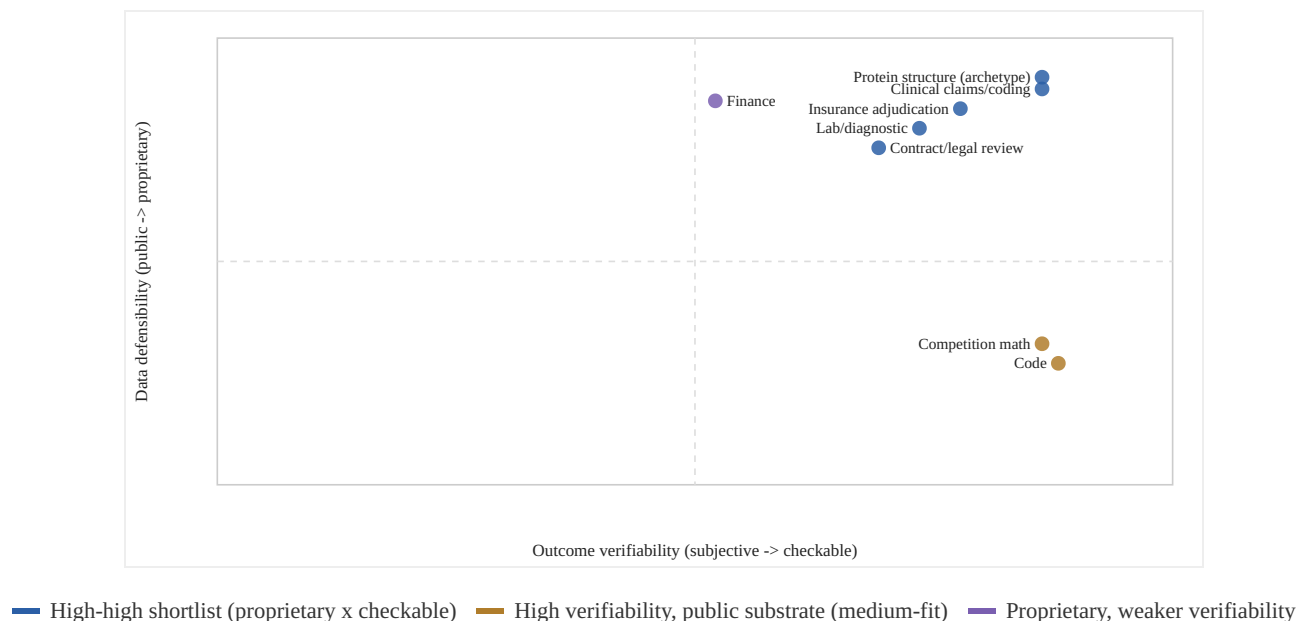


Figure 3. Vertical positioning: outcome verifiability versus data defensibility. A constructed 2x2 positioning of eight verticals on the two filter axes. Upper-right (high-high shortlist): clinical claims/coding, insurance adjudication, lab/diagnostic, contract/legal review, with protein structure (AlphaFold) as the calibrating archetype. Lower-right (high verifiability, public substrate, medium-fit): code and competition math - excellent for technique, thin as a moat. Upper-left (proprietary but weakly verifiable): finance. Coordinates are an analytical placement derived from the cited empirical anchors, NOT measured values; the anchors themselves (Figure 1, Table 1) are empirical and sourced. *Analytical placement by the authors from cited evidence: AlphaFold (Nature 2021, GDT_TS 92.4 CASP14), SWE-bench (arXiv:2310.06770), FrontierMath (arXiv:2411.04872), BloombergGPT (arXiv:2303.17564), MEDITRON (arXiv:2311.16079), Med-PaLM 2 (arXiv:2305.09617), LegalBench (arXiv:2308.11462). Constructed strategic analysis, not empirical data. · Constructed analysis.*

8. Limitations and Counterarguments

This scorecard ranks only data shape. It scores two of Perpetual Labs' seven relevant decision factors — clear reward signal and proprietary-data defensibility — and is silent on AI-automation alpha, vertical monopoly dynamics, AI-integration greenfield, market-entry ease, and competitor data-vendor gaps. A vertical can top the data-shape ranking and still fail the full go/no-go; the scorecard is the front of a funnel, not the decision.

The worked dental example carries binding constraints the filter does not capture. Data-resale and HIPAA rights govern whether a clinical corpus can legally become a transferable asset; single-clinic supply is small (on the order of ≈ 673 patients in the worked instance) and may not support true RL versus behavior cloning; and a behavior-cloning ceiling caps capability if the data reflects only existing human coding rather than verified-optimal coding. These are real and must be resolved separately from the data-shape score. Verifier design is itself a hard problem: even programmatic checkers can be gamed if the checkable proxy diverges from the true objective (e.g. a claim that adjudicates 'clean' but mis-codes the encounter), and 'correct' in a real clinical workflow is adjudicated, not computed.

On the evidence: benchmark numbers are reported from primary sources and vary by harness, prompting, and tool access. FrontierMath is reported only at its November 2024 launch point (all evaluated models under 2%) [9]; we do not headline later, secondary-aggregator reads of frontier scores, because they are not traceable to a primary source and any single late point is provisional. SWE-bench Verified figures are the developer's own reported results (Claude 2 1.96% at launch; Claude 3.5 Sonnet 33.4% and 49.0%), and the field has shifted toward harder, standardized variants (SWE-bench Pro) where scores are markedly lower [8,15]. The data-stock figure ($\sim 300T$ tokens, ~ 2028 median) is an Epoch AI projection sensitive to overtraining and data-efficiency assumptions, not a measurement [1]. The market-loss figures (\$36B, \$19.7–25.7B, 11.8%, \$40–118) are industry and association estimates, not audited totals [20,21,22]. None should be read as precise; they establish direction and order of magnitude, which is what the thesis requires.

Finally, the RLVR-elicitation counterargument cuts in our favor but must be stated honestly: if RLVR sometimes only surfaces latent skill [19], then environments in domains already saturated in pretraining may teach little. This strengthens the case for proprietary, under-documented verticals (where skill is genuinely absent) and against code/math as a data business — but it cautions against assuming every verifiable environment adds new capability.

9. Conclusion

The data wall and the rise of verifiable rewards jointly make proprietary verifiable-reward environments a manufacturable data asset class, with a dated window opening now [1,2,3]. The two-axis filter from insight i14 — proprietary data crossed with checkable outcomes — is the right selection lens, and running it produces a ranking rather than a principle. The verifiability axis is not assumed but demonstrated: across FrontierMath, SWE-bench, and MedQA, objectively checkable and refreshable benchmarks climb measurably while static aggregates like MMLU saturate.

High on both axes: clinical claims/coding adjudication, insurance adjudication, lab/diagnostic workflows, and contract/legal review — privileged corpora paired with deterministic outcomes [13,14,18], with AlphaFold (GDT_TS 92.4 at CASP14) the calibrating archetype [7]. Medium: code and competition mathematics, whose excellent verifiability is undercut by public data and elicitation-not-teaching dynamics [8,9,19]. The dental claims/coding bet is the worked high-fit example: a private clinical record crossed with a deterministic coding outcome, sitting on a multi-billion-dollar, per-claim verifiable error rate [20,21,22].

The scorecard is necessary but not sufficient. It nominates candidates by data shape; it does not clear the legal, supply, and capability-ceiling constraints that the full target decision must resolve. Used as the front of a funnel — data-shape filter first, full seven-factor lens second — it converts a finite-text crisis into a forward-looking map of where created data compounds into defensible value, and the timing is now.

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Insight i14 derived the right filter — proprietary data crossed with a checkable outcome — but never ran it; this paper runs it and produces a ranked target list. The verifiability axis is not assumed but demonstrated: FrontierMath ran from all evaluated models under 2% at its November 2024 launch to 40.3% (GPT-5.2 Thinking, Tier 1-3, Python tools) by Dec 2025, SWE-bench Verified climbed from Claude 2's 1.96% (2023) to Claude Opus 4.5's 80.9% (Nov 2025, the first model past 80%), and MedQA saturated as frontier reasoning models (Gemini 3 Pro, GPT-5.1, GPT-5.2) topped it — objectively checkable, refreshable benchmarks climb measurably while static aggregates like MMLU saturate above ~86%, and the contamination-resistant frontier (SWE-bench Pro held-out commercial set 17.8%; LiveMedBench best 39.2% with 84% of models degrading post-cutoff) confirms that durable signal needs proprietary substrate, not just a checker. Run the two axes and a shortlist falls out. HIGH ON BOTH (fund the quadrant): clinical claims/coding, insurance adjudication, lab/diagnostic, contract/legal review — privileged corpora paired with deterministic pass/fail outcomes, with AlphaFold (GDT_TS 92.4 at CASP14) as the calibrating archetype. MEDIUM (do not mistake for a moat): code and competition math, whose excellent verifiability is undercut by public corpora and by RLVR often eliciting rather than teaching. The worked high-fit example is dental claims/coding: a private clinical record adjudicated against deterministic CDT/ICD/X12 codes, sitting on a quantified, recurring market loss — ~\$36B/yr in coding errors, ~12% of claims mis-coded (AMA), an 11.8% initial denial rate in 2024, \$40-118 to rework each denial, and ~\$19.7-25.7B in provider denial/adjudication spend. The economics are software-shaped: the marginal unit of training signal is a program execution, not a human label. Timing is the edge — the data wall (~300T tokens, ~2028 median) arrives exactly as this asset class forms. CAVEAT: this scores only 2 of 7 factors (reward signal + data defensibility); it is silent on automation alpha, monopoly dynamics, integration greenfield, entry ease, and competitor gaps, and does not clear the dental bet's binding constraints (HIPAA/resale rights, single-clinic supply n approximately 673, behavior-cloning ceiling). Use it as the front of the funnel, not the go/no-go.

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